

Article

Toward Sustainable Geohazard Assessment: Dynamic Response and Failure Characteristics of Layered Rock Slopes Under Earthquakes via DEM Simulations

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Abstract

Understanding the dynamic response and failure mechanisms of rock slopes during earthquakes is crucial in sustainable geohazard prevention and mitigation engineering. The initiation of landslides involves complex interactions between seismic wave propagation, dynamic rock mass behavior, and crack network evolution, and these interactions are heavily influenced by the slope geometry, lithology, and structural parameters of the slope. However, systematic studies remain limited due to experimental challenges and the inherent variability of landslide scenarios. This study employs Discrete Element Method (DEM) modeling to comprehensively investigate how geological structure parameters control the dynamic amplification and deformation characteristic of typical bedding/anti-dip layered slopes consist of parallel distributed rock masses and joint faces, with calibrated mechanical properties. A soft-bond model (SBM) is utilized to accurately simulate the quasi-brittle rock behavior. Numerical results reveal distinct dynamic responses between bedding and anti-dip slopes, where local amplification zones (LAZs) act as seismic energy concentrators, while potential sliding zones (PSZs) exhibit hindering effects. Parametric analyses of strata dip angles and thicknesses identify a critical dip range where slope stability drastically decreases, highlighting high-risk configurations for earthquake-induced landslides. By linking the slope failure mechanism to seismic risk reduction strategies, this work provides practical guidelines for sustainable slope design and landslide mitigation in tectonically active regions.

Keywords: earthquake-induced landslide; bedding/anti-dip slopes; Discrete Element Method; sustainable geohazard assessment; rock slope stability; critical dip angle



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1. Introduction

Earthquake-induced landslides are infamous for their suddenness and severe destructive potential, causing building destruction, traffic blockages, and complications in post-earthquake rescue efforts [1–4], which are highly complex processes characterized by nonlinear dynamic problems, including amplification effect, fracture, collision, and shear sliding along an inclined plane. Researchers have invested considerable effort in understanding the dynamic response, deformation and failure mechanism of slopes subjected to seismic activity [5–8]. Based on previous data [9,10], the primary factor determining the slope's anti-sliding capacity is the mechanical properties of the stratigraphic lithology.

For instance, formations such as weak rocks, metamorphic rocks, marls, and clayey soils are more prone to slide, inducing massive collisions and friction during the movement process, which is usually accompanied by the strain-softening effect [9,11,12]. Furthermore, one of the internal factors contributing to landslides is geological structure, such as strata parameters and structural joint faces [13–15], where the rocks are fragmented, lack integrity, and have low strength, thereby providing the structural interfaces or sliding surfaces for landslide movement.

Among landslide hazards with various geological structures, typical bedding/anti-dip slopes, representing quintessential manifestations of stratified landslide geometries, exhibit the highest incidence. The landslide movement of bedding slope is identified as four stages [16]: the acceleration stage, steady movement stage, rapid movement stage, and deceleration and deposition stage. While the failure of anti-dip slopes containing weak layers can be divided into six stages [17]: slight damage to the slope, extrusion of soft rock, tensile fracture of hard rock on the interface, upward propagation of cracks in hard rock, sliding and extrusion of weak rock layers, sliding through the interconnection of bedding planes and weak rock layers, and slope failure. Generally, bedding slopes are more prone to sliding, whereas anti-dip slopes are relatively stable. The investigated dynamic mechanisms of landslides demonstrated that weak interlayers can amplify seismic waves, potentially accelerating landslide occurrences [18]. The dynamic amplification effect of local topography on the slope is evident for both bedding and anti-dip slope, mainly concentrated in areas with prominent slope topography, multiple free faces and slope shape transitions [19–23]. Bedding planes play a predominant role in slope failure, and both high-/low-frequency components of seismic waves contribute to the local deformation and overall sliding [24]. The deformation and failure characteristics of slopes under seismic conditions are also crucial for the mode and magnitude of such disasters. However, there is still insufficient understanding of the seismic amplification factor within slopes and the influence of geological structure parameters on the failure characteristic of landslides.

Currently, the mainstream approach for studying the earthquake-induced landslides is to use model testing to reproduce the movement and failure processes of landslides [16,18,25]. However, due to the constraints of experimental limitations, such as size and boundary constraints, time-consuming, laborious, and costly numerical simulation methods are extensively adopted to investigate the dynamic response of slopes [26,27], among which the Discrete Element Method (DEM) demonstrates particular advantages. A prerequisite in DEM modeling is to calibrate the material properties by customizing a set of micro-mechanical parameters for the particles and the contact bonds so as to yield desirable properties [28,29]. The parallel bond model (PBM) in the Particle Flow Code (PFC) is commonly used to replicate the mechanical properties of rocks. However, the resulting compression–tension strength ratio (CTSR) fluctuates only within the range of 3–7, which is lower than the CTSR of common rocks (10–20) [11,30,31]. Numerical models with a lower CTSR tend to exhibit larger errors when simulating problems involving both compressive and tensile stress paths, especially problems revolving around the deformation and failure of weak rock mass [28,32]. To better simulate the failure behaviors of the quasi-brittle type rocks [33], Ma and Huang [34] proposed a displacement-softening contact model, i.e., the soft bond model (SBM), modifying from the parallel bond option, which achieves a realistic CTSR as high as about 30 and captures the highly nonlinear failure envelope in the low confining stress range through adjusting bond strength ratio, softening factor, and particle shape [35]. “Displacement softening” essentially considers the progressive nature of micro-scale failure and increases the interaction range between particles in contact, and thereby is better able to capture the nonlinearity and failure process of quasi-brittle type material. Adopting SBM, the simulation on hybrid adhesive joints with dissimilar

adherend materials can characterize the transitional micro-mechanical response and failure mode accurately [36]. The rate-dependent strain-softening contact model based on the SBM framework has been utilized to reproduce the complete dynamic failure process of the Hongshiyuan seismic landslide, demonstrating both effectiveness and reliability in simulating earthquake-induced slope instability [37].

The workflow diagram of the methodology in this work is summarized in Figure 1. In this study, taking the generalized layered slope as an example [19], DEM models for bedding/anti-dip cases are established via PFC considering the strain-softening effect through reasonable geomechanical generalization, incorporating contact model, parameter calibration and boundary condition. The characteristics of the dynamic response of rock slopes and the fracture deformation subject to seismic excitation are investigated numerically in detail. A comparative analysis is performed on the dynamic response characteristics, failure patterns and temporal evolution of crack number for slopes with different geological structure parameters. The following discussion explains the features of dynamic response and deformation characteristic, along with the impact of structure parameters. The conclusions are summarized at the end.

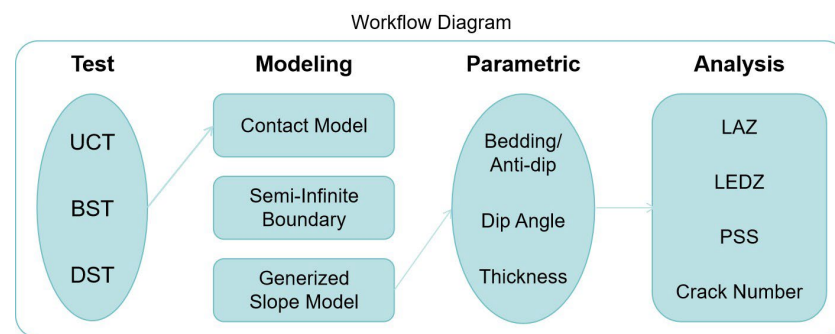


Figure 1. Workflow diagram illustrating the methodological framework implemented in this study (the abbreviations are explicitly clarified in the later sections).

2. DEM Modeling and Parameter Calibration

2.1. DEM Formulation and Mechanical Behavior

Layered rock slopes constitute complex geological media that involve features such as fissures, joints, faults, and bedding planes, which strongly affect the performance of engineering structures. To better elucidate the role of structural planes in the landslide process, we simplified the modeling of the layered slope as rock masses connected by mutually parallel joint planes, while disregarding other irregular rock parameters.

The PBM effectively transmits both force and moment at the contact point, with the shear force, bending, and twisting moment, and thus, it is widely used for simulating rock mass failure. However, PBM cannot accurately capture progressive softening due to internal fracturing [31]. Contrary to PBM, in which the bond is removed upon failure, the bond in SBM may enter a softening state until the bond stress reaches a threshold value before the bond is removed [34]. The SBM allows particles or blocks to transmit force and moment through a virtual “soft” connection, which can possess properties such as elasticity, plasticity, and fracture, thereby simulating the failure behavior of a large-sized granular rock mass. These settings ensure that the simulated rock mass adheres to the required CTSR from the experimental results, as depicted in Section 2.2. The normal stress (σ) and shear stress (τ_m) in the SBM are updated using the following equations:

$$\sigma = \frac{F_n}{A} + \beta \frac{||M_b||R}{I} \quad (1)$$

$$\tau_m = \frac{||F_s||}{A} \quad (2)$$

where F_n represents the normal force of the contact in N, A represents the cross-sectional area of the contact in m^2 , β represents the bending moment contribution coefficient (typically set to 1), M_b represents the bending moment of the contact in N·m, R represents the radius of the contact in m, and I is the moment of inertia of the contact, calculated as $I = \frac{2}{3}R^3$ in the 2D DEM simulation. F_s represents the shear stress of the contact in N. When the normal stress on the contact exceeds the bond strength, the contact transitions into a softened state. The maximum normal elongation (l^*) of the contact can be determined using the following equations:

$$l^* = l_c(1.0 + \zeta) \quad (3)$$

$$l_c = \frac{F_n}{k_n A} + \beta \frac{||M_b||R}{k_n I} \quad (4)$$

where ζ is the softening factor of the contact, k_n is the normal stiffness, and l_c is the elongation. The maximum normal elongation l^* corresponds to the peak stress σ^* of the contact:

$$\sigma^* = \frac{\sigma_c(l^* - l)}{\zeta l_c} \quad (5)$$

where l is the current elongation of the contact, and σ_c is the normal stress.

The presence of joint faces between rock strata affects failure characteristics, leading to various failure modes, which conversely influences the propagation of seismic waves and the dynamic response in slopes [38]. For constructing a refined slope DEM model, this study employs the Smooth-Joint Model (SJM) to simulate the mechanical behavior of the joint face. The SJM is commonly used to describe the contact interface between particles or between particles and boundaries, particularly when interfaces exhibit smooth joint characteristics, such as linear elastic behavior, bonded failure, non-bonded friction, and dilatant friction [39]. SJM can be visualized as a series of springs uniformly distributed along a circular cross-section, with the centers located at the contact points and their directions parallel to the joint surface. The joint surface behaves linearly, with the normal force F_n and the shear force F_s given by

$$F_n = k_n \delta_n, \quad F_s = k_s \delta_s \quad (6)$$

where k_n and k_s are the normal and shear stiffness coefficients, respectively; δ_n and δ_s are the normal and shear displacements, respectively. The shear force follows Coulomb's friction law: $|F_s| \leq \mu F_n$, where μ is the coefficient of friction. When the shear force reaches the bond strength limit, the bond breaks and the interface behavior is then governed by the friction. After generation of the joint plane, a smooth joint is assigned to the particle contacts with centers located on the opposite sides of the joint plane. At these contacts, the soft bonds are removed and smooth joints are defined in the direction parallel to the joint plane.

2.2. Selection and Calibration of Micro-Mechanical Parameters

To build a reliable DEM model that realistically reflects the mechanical behavior of rock masses and joint faces, the micro-mechanical properties of particles are distilled from the calibration process between laboratory experiments and numerical models to capture the response of the uniaxial compression test (UCT), Brazilian splitting test (BST), and direct shear tests (DSTs), as shown in Figure 1. The rock samples used in this study were obtained from large rock blocks with prominent structural surfaces, which were detached from the Tangjiashan landslide mass after the 2008 Wenchuan Earthquake. These quasi-brittle

limestones were subsequently transported to a cutting facility and processed to 10-cylinder samples of $\Phi 50 \text{ mm} \times 100 \text{ mm}$ for UCT (Figure 2a,b) and $\Phi 50 \text{ mm} \times 100 \text{ mm}$ for BST (Figure 2c,d). The UCTs and BSTs are conducted using a HCT206A computer-controlled electro-hydraulic servo rigid pressure testing machine at a loading rate of 0.5 MPa/s to obtain the stress–strain relationship of the samples. The DSTs are conducted on 5 cube rock samples ($50 \text{ mm} \times 50 \text{ mm} \times 50 \text{ mm}$) (Figure 2e,f) with a joint face using a YZW-30B microcomputer-controlled electronic rock direct shear apparatus at a loading rate of 1.0 MPa/s. Table 1 lists the experimental mechanical parameters of rock mass and joint face essential for the parametric calibration.



Figure 2. Limestone samples and failure states for UCTs (cylinder; (a,b)), BSTs (cylinder; (c,d)) and DSTs (cube; (e,f)). Google Earth-based map (g) of the location where the sampling blocks were collected.

Table 1. Experimental mechanical parameters of the rock mass and joint face.

Mechanical Parameters	Value
Compressive strength (MPa)	85.04
Elastic modulus (GPa)	12.23
Poisson's ratio	0.21
Tensile strength (MPa)	7.29
Cohesion of joint face (MPa)	0.35
Angle of internal friction of joint faces ($^{\circ}$)	36.00

Based on macroscopic observations and petrographic comparisons, the rocks can be classified as micritic to biosparitic limestones containing a significant proportion of muddy matrix. This lithology is consistent with the microbial mound-bearing muddy limestones identified in the Xiannüdong Formation in the Tangjiahe section of northern Sichuan. Specifically, the limestones in this region frequently contain microbial fossils such as Epiphyton and Renalcis, with a matrix of recrystallized micrite and minor detrital quartz, glauconite, and fossil fragments including trilobites and brachiopods, indicating deposition in a warm-water carbonate slope to a platform-margin setting. The sampling site is located near Tangjiashan (Figure 2), situated on the right bank of the Tongkou River approximately 6.5 km upstream of Beichuan County, Sichuan Province ($104^{\circ}25'56.93'' \text{ E}$, $31^{\circ}50'40.60'' \text{ N}$). The site lies in the southwestern part of the Micang Mountains and is part of the upper Yangtze Platform's northern margin; during the Early Cambrian, it was characterized by a passive continental margin setting with alternating carbonate and clastic deposition. The formation exhibits complex lithological assemblages including stromatolitic limestones, intraclastic rudstones, and micritic limestones rich in algal structures. These lithofacies are closely associated with slope collapse and reworking processes, consistent with the landslide-induced geological deformations observed at the Tangjiashan site. Structurally, the region is heavily influenced by the Longmenshan Fault Zone. The Tangjiashan landslide mass is situated within the hanging

wall of the central Longmenshan fault (also known as the Beichuan–Yingxiu fault), in proximity to the overturned Qinglinkou anticline. The rock mass is characterized by steeply dipping bedding planes and highly developed joint sets, leading to reduced rock mass integrity and a predisposition to large-scale landslide failures [40].

For micro-parameter calibration of rock mass, a 2D rectangular vessel (50 mm × 100 mm) consisting of frictionless walls with an assembly of grains satisfying a uniform size distribution (0.4–1.6 mm) is constructed to simulate the rock sample for UCT. The grains are rearranged into a static-equilibrium state under zero-friction conditions and the soft bonds are then introduced. Vertical compression force on the top is applied by activating the servomechanism with a pressure boundary condition. With the same setting of particles, a 2D circular vessel ($\Phi 50$ mm) is constructed to simulate the rock sample for BST. Vertical force is applied on the platen tangent to the top of the circular region, activating the servomechanism with a force boundary condition, and the walls on the circular boundary are set free. The effective modulus (E) and normal-shear stiffness ratio (k^*) are established through matching the experimental mechanical elastic modulus and Poisson’s ratio by UCT. The softening factor (ζ) and bond strength ratio ($K_r = \sigma_n/\sigma_s$) are then determined by reproducing the CTSR by UCT and BST. Following a multiple iteration process, five micro-parameters of SBM are calibrated. The computed stress–strain response is compared with that measured in the laboratory for intact rocks under UCT and BST, as depicted in Figure 3a–c. The numerically simulated strengths closely match the laboratory-tested values, indicating that the calibrated micro-mechanical strength parameters accurately reflect the rock’s mechanical behavior. Furthermore, the CTSR obtained from laboratory tests (11.66) and numerical simulations (10.93) are close, enhancing the representative of the simulation in capturing actual mechanical behavior and failure characteristics. The failure snapshots of specimens for laboratory test and numerical simulation are shown in Figure 2b,d. The rock mass under uniaxial compression demonstrates characteristic brittle fracture behavior dominated by tensile crack propagation, a phenomenon accurately replicated in the DEM simulation, which showed a predominantly tensile failure mode. The BST result also shows good agreement between the numerical simulation and the laboratory test. It can be concluded that the micro-properties obtained using the described calibration process ensure that the model reasonably predicts the response of the rock mass. The calibrated values of the micro-mechanical parameters for rock mass are listed in Table 2.

Table 2. Micro-mechanical parameters of the rock mass and joint face.

Micro-Mechanical Parameters	Description	Value
Rock mass		
E	Effective modulus (GPa)	32.0
k^*	Stiffness ratio	3.5
σ_n	Normal strength (MPa)	18.0
σ_s	Shear strength (MPa)	70.0
ζ	Softening factor	20.0
Joint faces		
k_n^{sj}	Normal stiffness (N/m)	1.2×10^{11}
k_s^{sj}	Shear stiffness (N/m)	3.2×10^{10}
σ_n^{sj}	Tensile strength (MPa)	6.0
c^{sj}	Joint cohesion (MPa)	7.9
μ^{sj}	Joint friction	0.7
ψ^{sj}	Dilation angle (°)	60.0

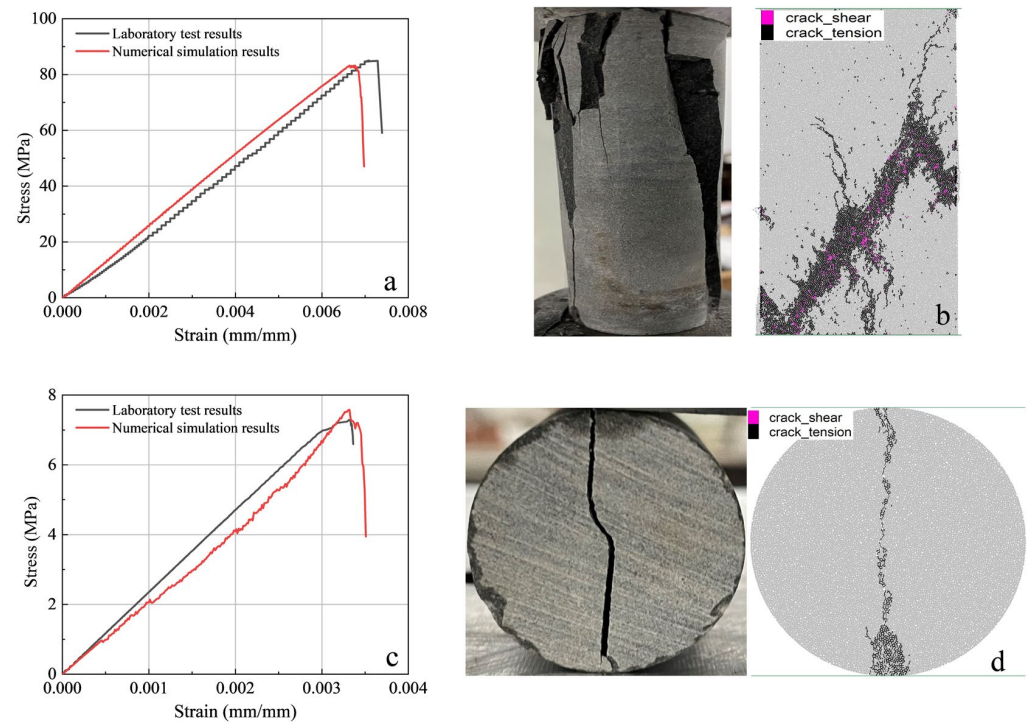


Figure 3. Stress–strain curves obtained from the UCT (a) and BST (c) and damage characteristics of uniaxial compression (b) and Brazilian splitting. (d) Specimens from laboratory tests and DEM simulations.

Based on the rock mass model above, a model ($50 \text{ mm} \times 50 \text{ mm}$) for the structural plane with joint face is established. A search function is used to set contacts at the interface between the two groups of particles as the structural plane section bonded by SJM, while contact between the other particles is achieved using SBM [36]. The corresponding micro-properties for the SJM have six essential parameters: joint normal stiffness (k_n^{sj}), shear stiffness (k_s^{sj}), joint tensile strength (σ_n^{sj}), joint cohesion (c^{sj}), joint friction (μ^{sj}), and dilation angle (ψ^{sj}). The values of joint normal stiffness (k_n^{sj}) and shear stiffness (k_s^{sj}) solely depend on macroscopic normal stiffness K_n and shear stiffness K_s , respectively. The calibration procedure is as follows: get μ^{sj} based on the shear stress on the structural plane; calculate k_n^{sj} and k_s^{sj} based on K_n and K_s ; determine ψ^{sj} through the ψ of the structural plane using the Mohr-Coulomb criterion; and search c^{sj} using a multiple iteration process. Figure 4a shows that the maximum shear force increases with normal stress but drops almost instantaneously once the peak is reached. The shear stress can be expressed as $\tau = c + \sigma_n + \tan\varphi$, where τ is the shear stress, c is the cohesion, σ_n is the normal stress, and $\tan\varphi$ is the angle of internal friction. The results from both laboratory tests and DEM simulations are fitted, as illustrated in Figure 4b. The shear stress of the DEM specimen closely aligns with the laboratory test results, indicating that the SJM with calibrated parameters effectively captures the shear mechanical behavior of the joint face. The calibrated values of the micro-mechanical parameters for the joint face are listed in Table 2.

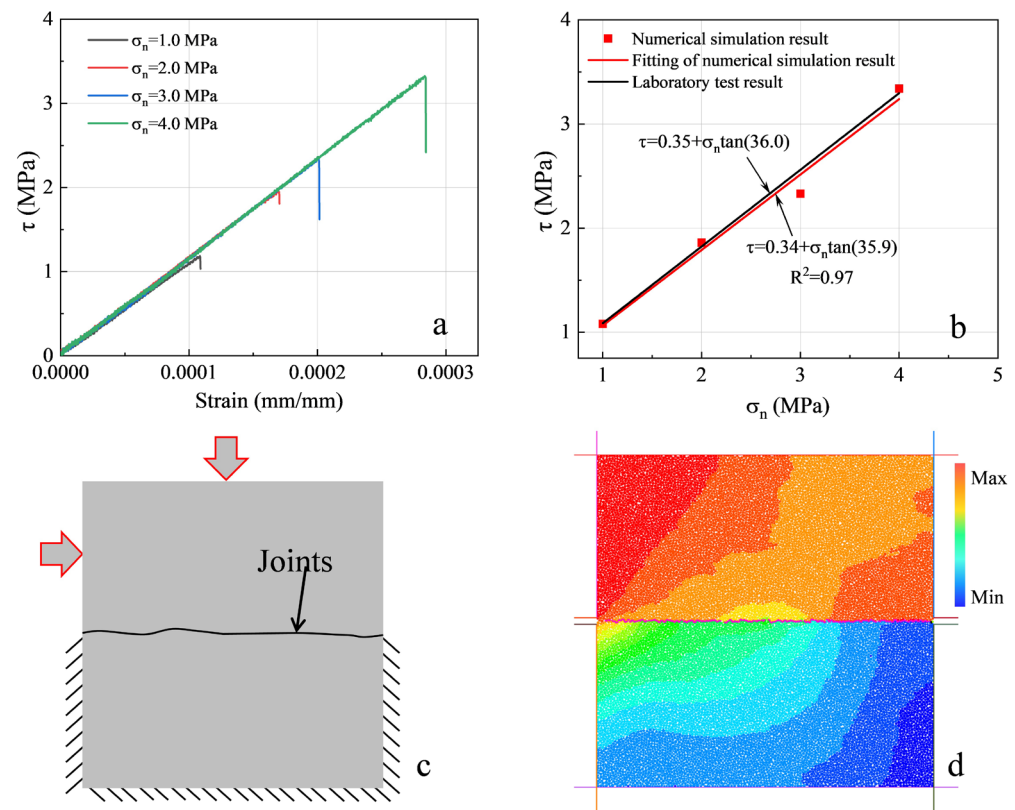


Figure 4. The calibration of micro-mechanical parameters of the joint face. (a) Experimental results of DST with different normal stresses, (b) calibration results, (c) frame of the numerical model with SJM, (d) displacement field by the DST simulation.

2.3. Boundary Conditions

Seismic waves propagated in granular media are reflected when they encounter a free surface. However, the boundaries outside of the slope, aside from the free surfaces, act as semi-infinite media without reflection. To address this, artificial boundaries are used to absorb the energy of the incident wave. The propagation velocity (V) of seismic waves in a continuous medium is related to the elastic modulus E and density ρ : $V = \sqrt{E/\rho}$. When a seismic wave reaches the semi-infinite boundary of a slope, a counteracting force is required to neutralize its effect. The relationship between the opposing contact force (F_b) on the semi-infinite boundary and the velocity of particle motion (v) can be calculated as follows:

$$F_b = -2R\rho Vv \quad (7)$$

To evaluate the effectiveness of the semi-infinite boundary in absorbing seismic waves, a wave propagation test is conducted using the model ($120 \text{ m} \times 60 \text{ m}$) shown in Figure 5a. The seismic waveform monitored at points 1 and 2, as shown in Figure 5b, aligns with the incident waves and exhibits no reflections. The wave velocity is calculated based on the timing of the wavefront's arrival at points 1 and 2 is 3600.5 m/s , which closely matches the theoretical value for rock mass. Due to the attenuation of seismic waves, the waves monitored at points 1 and 2 are slightly smaller than the incident waves. The velocity field in Figure 5c(1–3) also confirms that the seismic waves propagate without reflection.

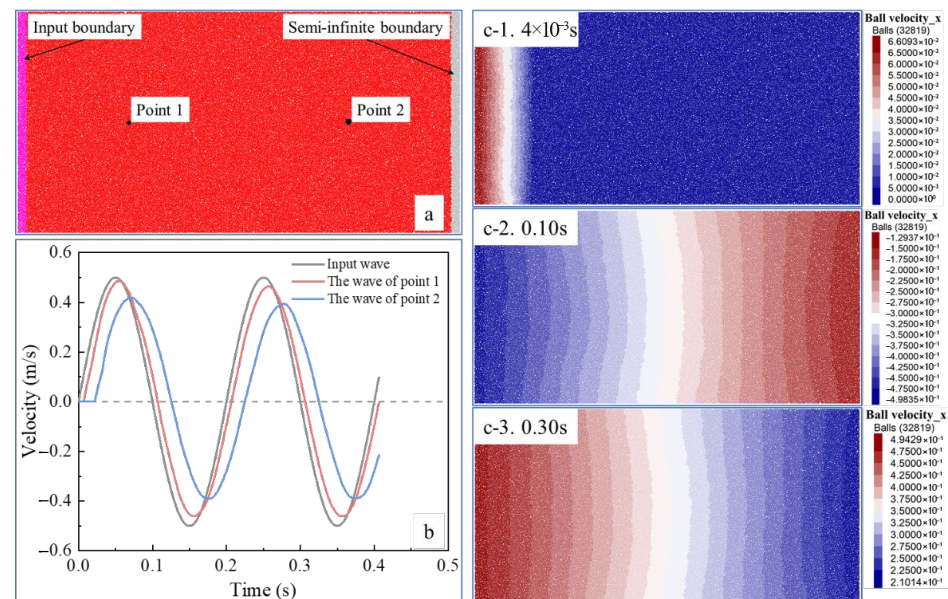


Figure 5. Test results for a semi-infinite boundary. (a) Testing model; (b) velocity of particles at point 1 and 2; ((c)1–3) velocity field of particles at three different times.

3. Simulation on Layered Slope Under Earthquake

3.1. DEM Model of the Generalized Layer Slope

A DEM model of the typical bedding/anti-bed layered rock slope is established via PFC with the micro-mechanical parameters calibrated above. The relative elevation of the slope is 120 m, the length is 175 m, and the gradient is 50° , as shown in Figure 6a. The left and right boundaries of the model are treated as semi-infinite. The base of the slope serves as the boundary for vertical seismic wave input (Figure 6b), with a velocity time history of the Wolong wave [41]. To reduce computational cost, the signal is calibrated and only the seismic wave data within the red box (Figure 6c) is chosen for application. The EQ toolbox is used to perform baseline correction on the seismic waves based on the principle of the least squares method. The input signal contains a total of 50 s of seismic process, featuring two prominent peaks, each representing an acceleration peak, occurring at the time points of 13 s and 33 s. The velocity ratio is defined as $V_r = v(t)/V(t)$, where $v(t)$ is the real-time velocity of particles in the slope and $V(t)$ is the input seismic wave velocity. The simulation result of velocity ratio and displacement are monitored at 10 s, 20 s, 30 s, 40 s, and 50 s (as marked in Figure 6d). The study was conducted over a duration of 28 h for each case using a workstation equipped with Intel(R) Core (TM) CPU i7-6950X @ 3.00 GHz processors and 128 GB of RAM.

3.2. Dynamic Response and Deformation Characteristics of Bedding/Anti-Dip Slopes

The dynamic response and displacement process of bedding/anti-dip layered slopes ($h = 10$ m, $\theta_{dip} = \pm 35^\circ$) are illustrated in Figure 7. For the bedding case, the heterogeneous distributed dynamic response exhibits values of V_r approaching 2.5 in most locations at 20 s, indicating a significant amplification effect. An interesting observation is the occurrence of V_r exceeding 1.25 within the rock mass during the second seismic peak stage (30–40 s), which gradually migrates from the inner part (30 s) towards the middle-to-upper surface of the slope and becomes more concentrated (40 s). This area is highlighted as the local amplification zone (LAZ, enclosed by a green curve), which could potentially be the primary excitation source of rockfall in slope. This migration indicates the reflection and collision effects resulting from the generated numerous micro-cracks and crack networks by earthquake. At 50 s, the amplified seismic wave within the slope dissipates, accompanied by the

loosening of the slope. For the deformation of the slope, it is evident that from 20 s to 50 s, the upper part of the slope has developed an interconnected local extreme-displacement zone (LEDZ, enclosed by a yellow dashed curve), also identified as the potential slip zone (PSZ), where fractured rock masses are prone to large sliding displacements, triggering landslides. The boundaries of LEDZ within the slope are composed of crack networks, encompassing both inter-layer and intra-rock cracks. The interconnected cracks that penetrate in the same direction form a major potential slip surface (PSS, red dashed-dotted line) and parallel minor PSSs (red dashed line). The slope remains highly hazardous due to extensive coverage with numerous fracture surfaces, constantly posing a risk of rockslides.

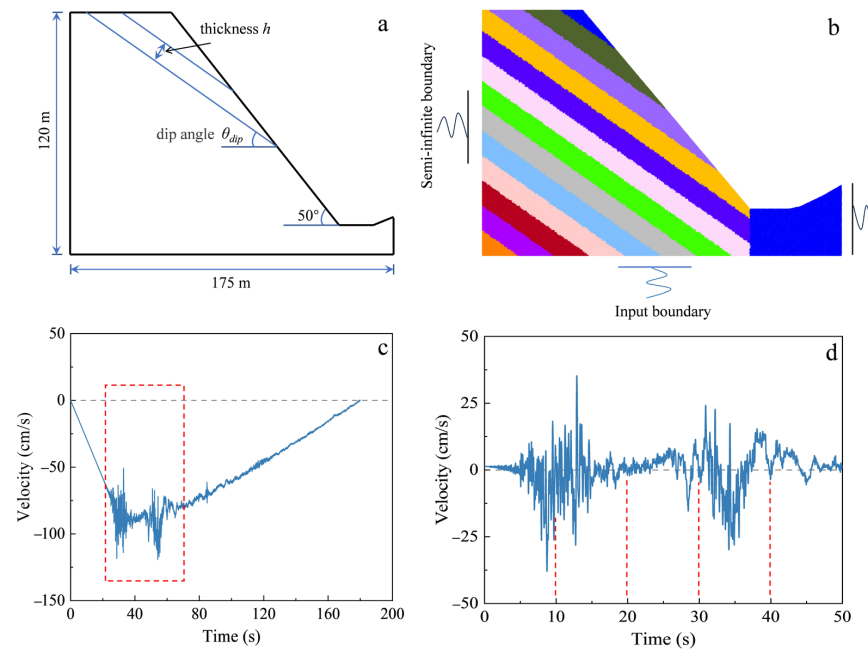


Figure 6. DEM model of generalized layered slope. (a) Model geometry. (b) DEM model under earthquake waves. (c) Original data of Wulong waves. (d) Calibrated curve.

The dynamic response observed in the anti-dip scenario is comparable to that observed in the bedding case. However, the cracks demonstrate a notably distinct pattern. During 20–50 s, similar amplification effects and dissipation processes are observed, albeit with minor local difference. The LEDZ for the anti-dip slope is a narrow zone comprising fractured blocks along the free surface with weak inter-layer connections, and the stability hinges on the overall integrity of these blocks. Despite the appearance of numerous penetrating cracks within the rock mass and LEDZ enclosed by a connected jagged sliding surface near the slope surface, the overall stability remains relatively well-maintained. The key block (cyan square) located beneath the bottom of the LEDZ, which bears the maximum compression load, plays a crucial role in maintaining the stability of anti-dip slopes.

As shown in Figure 8, the total velocity ratio of the bedding slope ($V_{r_tot} = \oint_S V_r ds$) reaches its maximum at the first seismic peak, not at the second one, due to the dissipation effects induced by the generated numerous cracks. Conversely, for the anti-dip slope, the larger V_{r_tot} during 30–40 s implies a higher degree of structural integrity after the destructive event than that of the bedding case. Therefore, during the cessation phase (40–50 s), the amplification effect ($V_{r_amp} = \oint_{S_{amp}} V_r ds$, $S_{amp} = S_{(V_r > 2)}$) of the anti-dip slope is larger than that of the bedding case, as shown in Figure 8a,b. The total displacement ($u_{tot} = \oint_S |u| ds$) for both the bedding and anti-dip slope exhibits two similar increase stages, coinciding with two seismic peaks. Although amplitudes of the LEDZ ($u_{LEDZ} = \oint_{S_{LEDZ}} |u| ds$, $S_{LEDZ} = S_{|u| > 0.3}$) exhibit similar trends in terms of the time domain,

the values of u_{LEDZ} for the bedding case are much larger than that of anti-dip case at the cessation stage. The distributions of LAZ and LEDZ at 40 s are also different for bedding and anti-dip slopes (Figure 8e,f), implying different slope failure modes, which are attributed to the directions of joint surface and the relative displacement along the sliding surfaces.

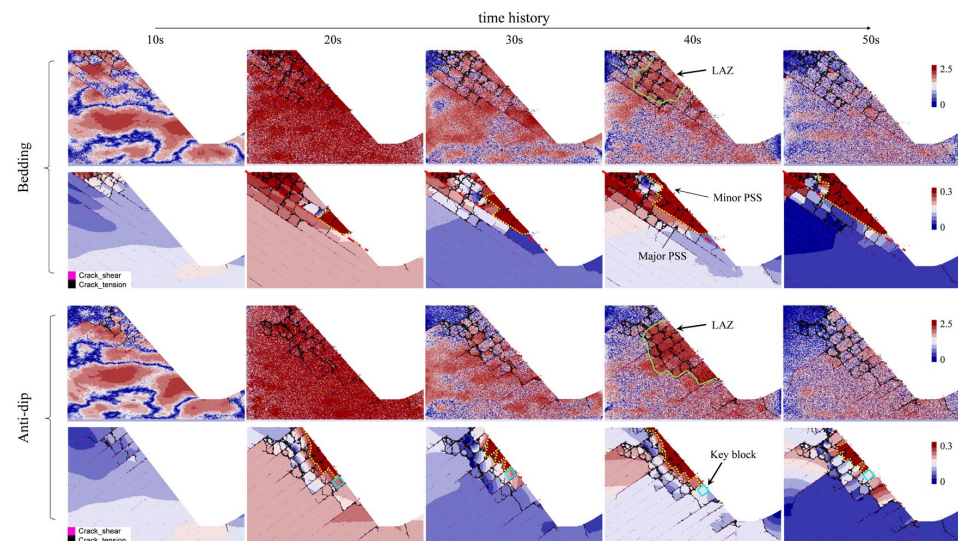


Figure 7. Dynamic response and deformation process of layered slope model for bedding/anti-dip slope cases ($h = 10$ m, $\theta_{dip} = \pm 35^\circ$) during 10–50 s.

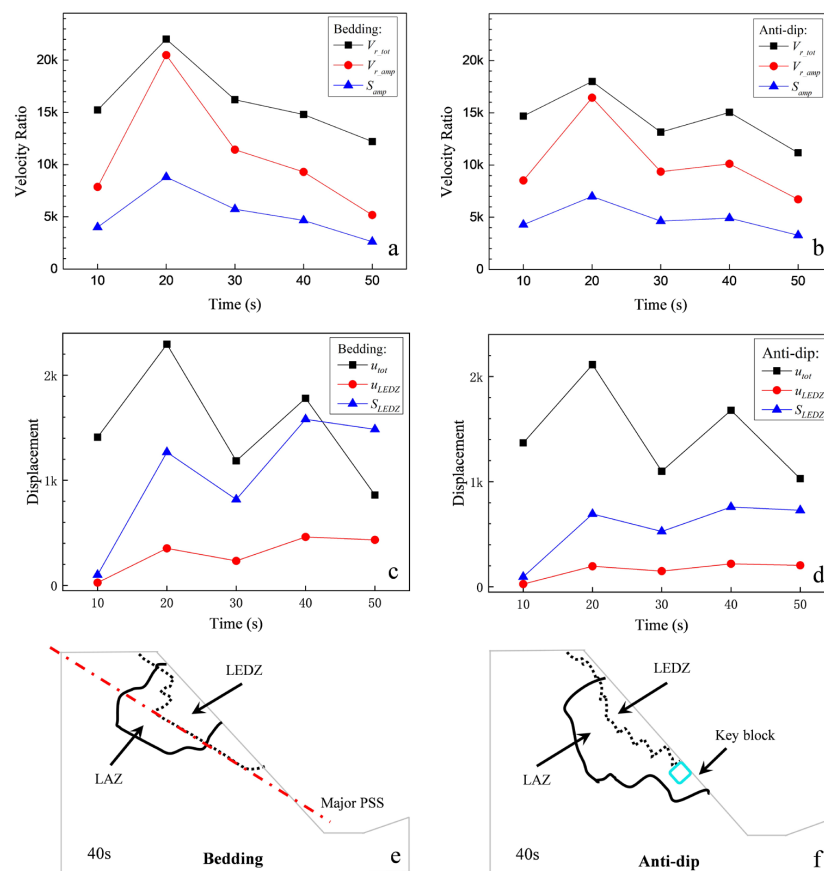


Figure 8. The amplification effect and displacement characteristic of bedding/anti-dip slope ($h = 10$ m, $\theta_{dip} = \pm 35^\circ$). Velocity ratio of the whole region (V_{r_tot}), the amplification effect (V_{r_amp}) and its area (S_{amp}) (a,b); displacement of the whole region (u_{r_tot}), LEDZ (u_{LEDZ}) and its area (S_{LEDZ}) (c,d); the distribution of LAZ and LEDZ for bedding/anti-dip slope at 40 s (e,f).

4. The Influence of Strata Parameters on Landslides

To investigate the impact of strata parameters, i.e., the strata dip angle θ_{dip} and strata thickness h , on the dynamic response and deformation characteristics of slopes, particularly the LAZ and LEDZ, a parametric study is conducted on the layered slope model. Computational models for bedding ($\theta_{dip} > 0$) and anti-dip ($\theta_{dip} < 0$) cases with varying θ_{dip} are constructed, including four types of slopes: flat layered ($\pm 5^\circ$), gently dipping ($\pm 20^\circ$), moderately dipping ($\pm 35^\circ$), and steeply dipping ($\pm 50^\circ$) slopes. For the moderately dipping slopes, models are created with strata thicknesses of $h = 5$ m, 10 m, 15 m, and 20 m.

4.1. Bedding/Anti-Dip Slopes with Different Strata Dip Angles

The simulation results of slopes with different θ_{dip} are observed in the post-earthquake phase ($t = 40$ s). As shown in Figure 9, the variations in the dynamic response of the bedding slope with θ_{dip} mainly focus on two aspects: intensity and distribution. For the flat layered slope, the LAZ is nearer to the middle of the slope and less distinct. Cracks are generated, developing into a fracture network with shear cracks serving as the primary structure and tensile cracks as branches, while a continuous and penetrating sliding surface is incompletely formed. For the gently dipping slope, the LAZ becomes more concentrated and prominent, connected with the slope surface, leading to severe cracking and failure. Although multiple PSSs are formed, none of them are fully connected in a unified direction, suggesting that the slope is stable overall. For the moderately dipping slope ($\theta_{dip} = 35^\circ$), the LAZ surrounded by cracked rock mass segments spreads across the upper part of the slope surface and a fully connected PSS is formed, as shown in the displacement field (and LEDZ marked by dashed curve) in Figure 9 (Bedding, 35°), which indicates a high probability of landslide occurrence. The activation of the landslide primarily depends on the friction of PSS and the extrusion force beneath the LEDZ near the surface of the slope. For steeply dipping slope ($\theta_{dip} = 50^\circ$), the LAZ is larger and more significant. A vast number of penetrating cracks are generated and cracks along the joint surfaces interconnect to form multiple PSSs. However, due to the support provided by the block at the toe, the overall slope maintains relatively stable.

The LAZ for anti-dip cases with varying θ_{dip} exhibit a pattern similar to that of bedding dip cases. As θ_{dip} increases, the amplification effect of seismic waves becomes more prominent and the LAZ expands from the intermediate position to the larger part near the free surface. Despite the localized fragmentation of the rock mass near the slope surface, the overall stability of the slopes remains relatively good for all dip angles. It is important to note that the key block at the toe of LEDZ for $\theta_{dip} = 50^\circ$ may suffer damage under upper dynamic loading.

The occurrence of micro-cracks is indispensable condition for initiating localized slip displacement. As θ_{dip} increases, the crack number steadily escalates, with initial cracks appearing at 8 s (Figure 10e,f). Notably, tension cracks vastly outnumber shear cracks in these slopes. Both type of cracks undergoes two phases of rapid growth, aligning with the seismic wave intensity. During the first phase (10–20 s), a remarkable increase in both tension and shear cracks appears, while the increase in shear cracks due to strata failure is relatively subdued and the proliferation of tension cracks resulting from failure in rock mass is more pronounced during the second phase (35–40 s). This observation implies that the shear cracks undergo a period of maturation after an initial seismic peak. Interestingly, the number of tension crack increases dramatically as the strata dips gently to moderately (20 – 35°), especially in anti-dip slopes. The significant rise in tension cracks within this θ_{dip} range indicates a heightened risk of high-toss action. The observed increase of u_{LEDZ} (S_{LEDZ}) within this θ_{dip} range for bedding cases (Figure 10c) also supports this inference. A similar tendency is also observed for V_{r_LAZ} (S_{LAZ}) in anti-dip case and u_{LEDZ} (S_{LEDZ})

in bedding case, as shown in Figure 10b,c, where V_{r_LAZ} is the value of velocity ratio in LAZ ($V_{r_LAZ} = \oint_{S_{LAZ}} V_r ds$, $S_{LAZ} = S_{(V_r > 2)}$). However, the displacement of LEDZ (u_{LEDZ}) in anti-dip case (Figure 10d) is inconsistent with the sharp rise of the crack number at 40 s. The main difference is that the bedding slope produces continuous and smooth cracks, while the anti-dip slope primarily produces zigzag-type crack networks, which might not necessarily contribute to u_{LEDZ} .

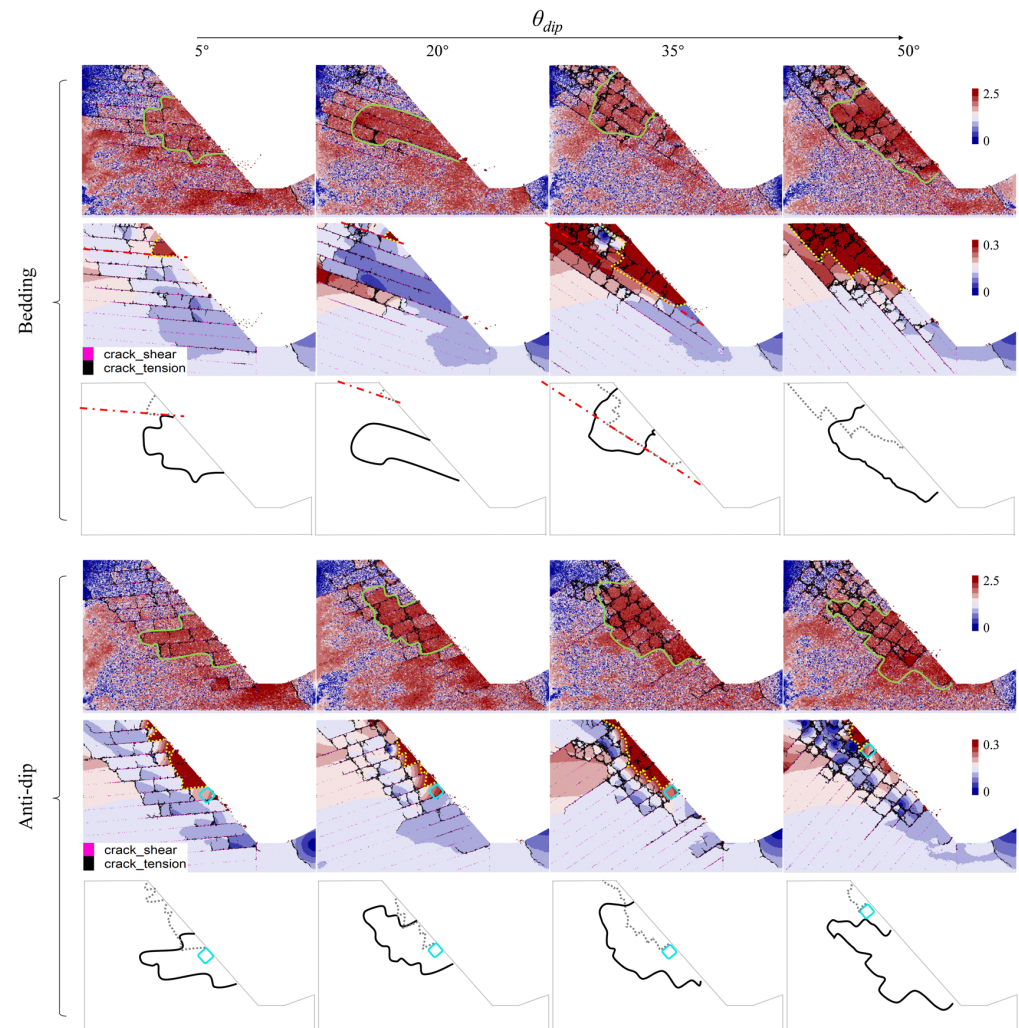


Figure 9. Dynamic response, deformation and distribution of LAZ and LEDZ in bedding/anti slopes ($t = 40$ s, $h = 10$ m) with different θ_{dip} ($\pm 5^\circ$, $\pm 20^\circ$, $\pm 35^\circ$, $\pm 50^\circ$).

4.2. Bedding/Anti-Dip Slopes with Different Strata Thicknesses

The distribution of dynamic response across slopes follows a unified pattern with different values for strata thickness h : a pronounced amplification zone concentrated in the upper parts of the slope surface for bedding case and in the middle parts for the anti-dip case, as shown in Figure 11. This observation suggests that the overall dynamic response and its distribution have little correlations with strata thickness h . However, the value of V_{r_LAZ} undergoes significant variations, especially for the anti-dip case. For bedding slopes, when $h = 20$ m, the amplification effect becomes more pronounced, which is likely attributable to the convergence between the layered slope's natural frequency and the dominant frequency of the seismic input. A similar mechanism can explain the fierce dynamic response observed in the anti-dip slopes with certain thickness values ($h = 5$ m, 10 m). The failure patterns of slopes with different h values exhibit similar distributions, while they are distinctly different for the bedding and anti-dip cases. A major PSS is

formed and fully connected along the direction of the joint surface for the bedding case, while connected and zigzag penetrating cracks develop, enclosing the LEDZs, and a root-like crack network is widely distributed within the slope for the anti-dip case. The total displacement u_{tot} values are comparable, while the u_{LEDZ} for the anti-dip case is much smaller than that of the bedding case (Figure 12c,d), which is consistent with the observation that the anti-dip slope holds better overall stability than the bedding slope. The numbers of cracks undergo two distinct phases of growth, corresponding to the time period of seismic wave peaks (Figure 12e,f). As strata thickness h increases, the shear and tensile cracks in both bedding and anti-dip slopes decreases uniformly, indicating that the integrity of the rock mass is enhanced. The consistent trend between u_{LEDZ} and crack number implies that the intra-rock and inter-layer cracks developed on the slope have a detrimental impact on the integrity of the slope.

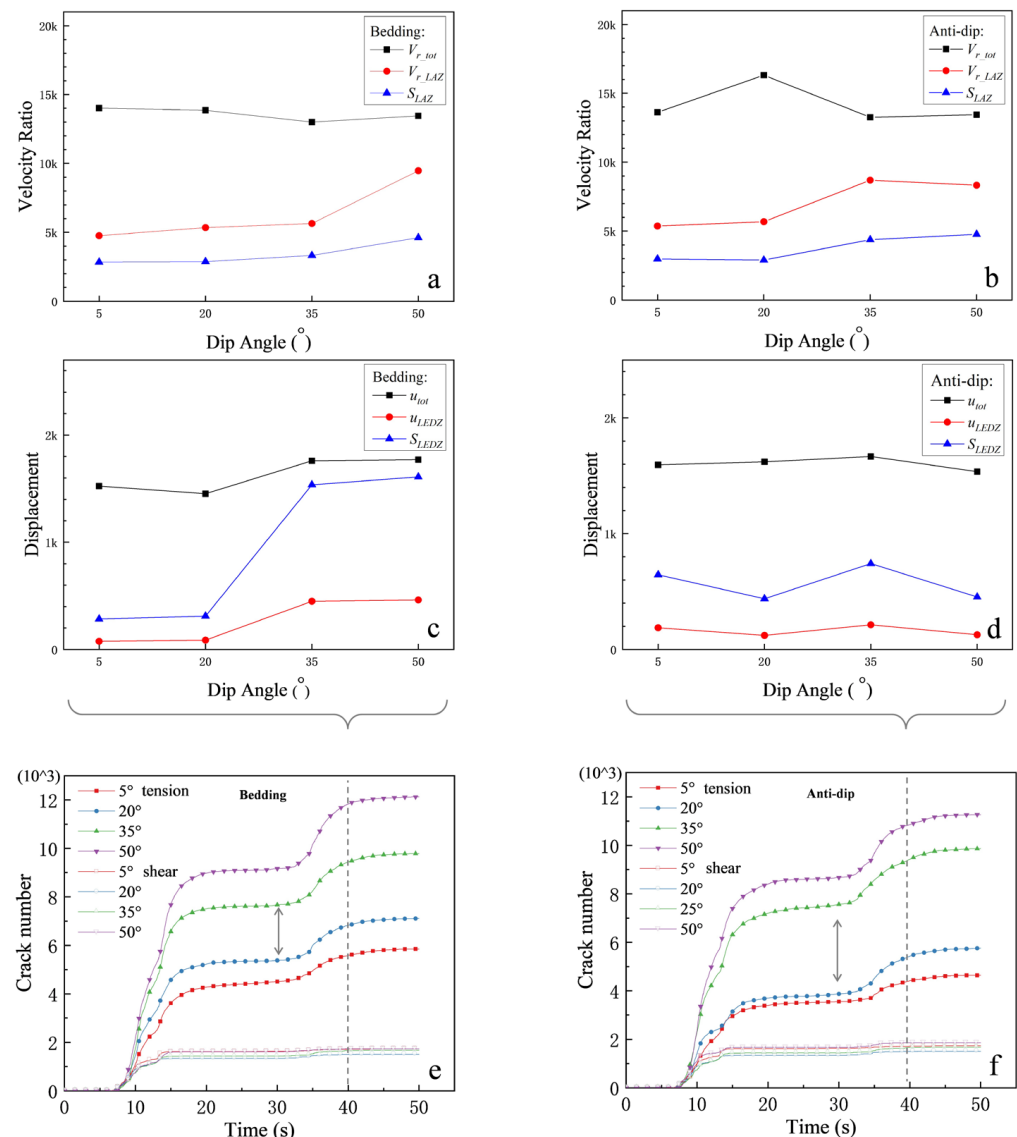


Figure 10. The velocity ratio of LAZ (a,b), displacement of LEDZ (c,d) at 40 s, and evolution of crack number (e,f) of bedding/anti-dip slopes ($h = 10$ m) with different θ_{dip} .

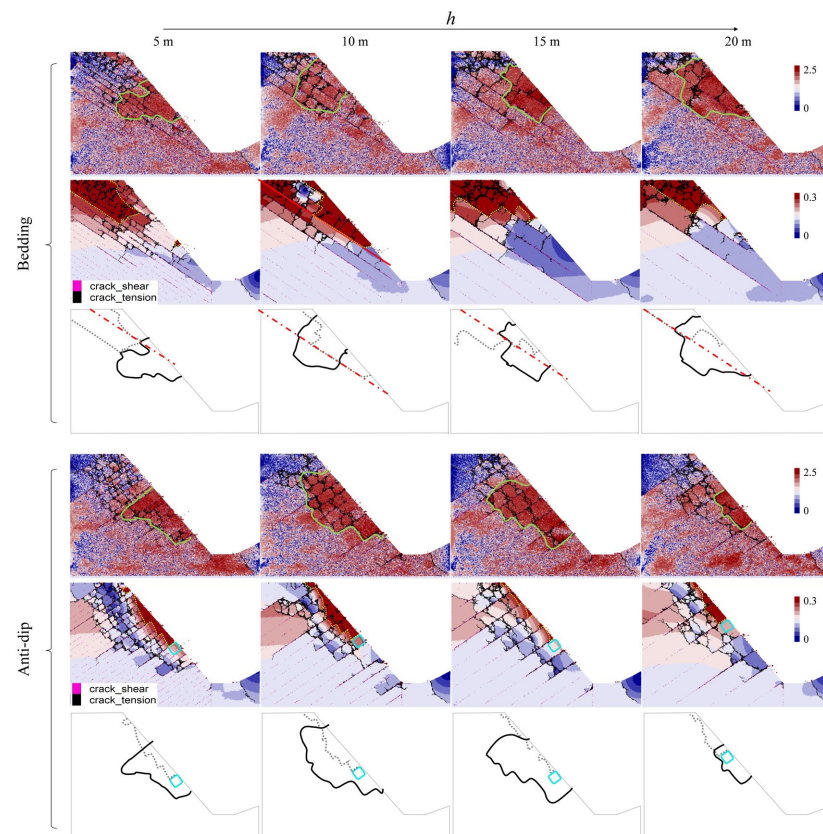


Figure 11. Dynamic response, deformation, and distribution of LAZ and LEDZ in bedding/anti-dip slopes ($\theta_{dip} = \pm 35^\circ$) with different h (5 m, 10 m, 15 m, 20 m).

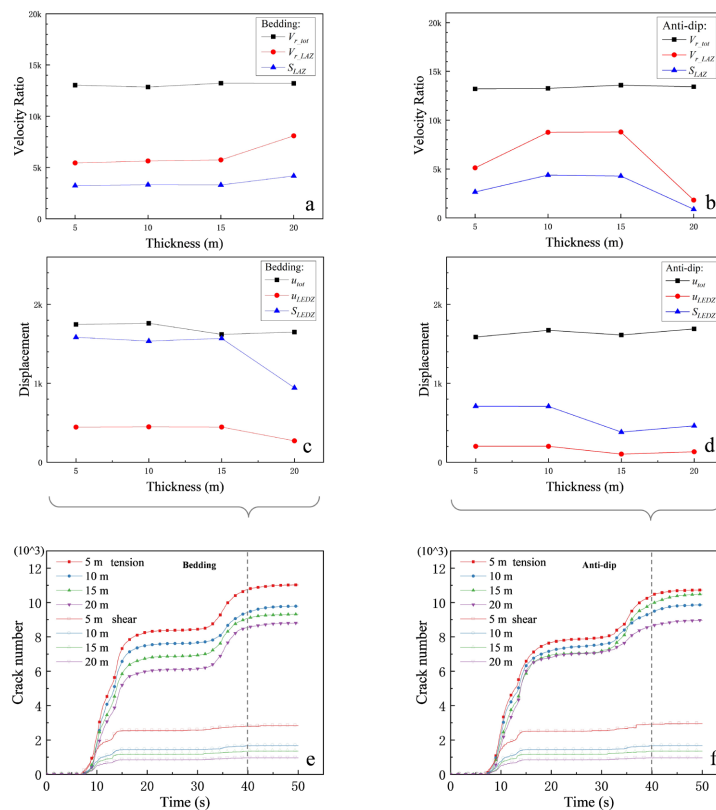


Figure 12. The velocity ratio of LAZ (a,b), displacement of LEDZ (c,d) at 40 s, and evolution of crack number (e,f) in bedding/anti-dip slopes ($\theta_{dip} = \pm 35^\circ$) with different h values.

5. Discussion

The focus of this work is on the dynamic response and failure mechanisms of layered bedding/anti-bed slopes in response to earthquakes. We simulate the dynamic response and deformation process of slopes under seismic loading in the time domain. At the initial stage, the slope exhibits an overall dynamic amplification effect (20 s), then transitions to localized amplification internally and gradually migrates from the inner part (30 s) towards the middle-to-upper surface of the slope and becomes more concentrated. The amplification phenomena near the free surface of the slopes are attributed to the reflection effect of seismic waves. At the cessation stage (40 s), crack networks give rise to LAZ and PSZ, which function as excitation sources and have a hindering effect, respectively. The interaction between those two zones contributes to the initiation of landslides. The slope experiences the progression from localized inter-layer micro-cracks to internal rock cracks, ultimately forming interconnected crack networks, among which the major through-cracks constitute the PSS. Luo et al. developed a novel analytical model for reactive contaminant transport in granular media, simultaneously characterizing boundary/internal sources and energy transfer [42]. This framework advances our understanding of contaminant migration in soils and tailings, though the potential geotechnical applications to slope dynamics require further study.

The influence of the distribution type of the strata, e.g., bedding and anti-dip, on the dynamic amplification effect and failure mode was investigated. The bedding slope is more likely to fail first, leading to the formation of cracks and dissipation, which in turn reduces the dynamic response (Figure 7). Due to the degree and mode of failure in the slope, the dynamic response of the anti-dip slope is greater than that of the bedding slope at the cessation stage (Figure 8a,b), while the stability of the anti-dip slope is better than that of the bedding slope (Figure 8e,f). These findings are consistent with previous field test results, as detailed in [17,43]. The interconnected crack network in the bedding slope, encompassing both inter-layer and intra-rock cracks penetrating in the same direction, creates a significant major PSS and parallel minor PSSs. An approximately triangular LEDZ containing intact rock layers is formed near the free surface at the top of the slope, which remains highly hazardous due to its extensive coverage with numerous fracture surfaces, posing a constant risk of rockslides. The LEDZ in the anti-dip slope exhibits a distinctly different pattern, featuring a narrow zone comprising fractured blocks along the free surface with weak inter-layer connections, and the stability hinges on the mechanical properties of its structure. The simulation results of the amplification effect and deformation characteristic with a dip angle of 50° is consistent with the shaking table test performed by Yang et al. on an anti-dip slope [44]. Despite the appearance of numerous penetrating cracks within the rock mass, the displacement of LEDZ is relatively small (Figure 8c,d) and the overall stability remains relatively well-maintained. The key block located beneath the bottom of the LEDZ plays a crucial role in maintaining the stability of the anti-dip slope, whose degradation may lead to large-scale catastrophic landslides, as observed in the Nayong rockslide in Guizhou Province [45]. Apart from the key block beneath the LEDZ, the integrity of the LEDZ is another important factor that determines the stability of anti-dip slope, necessitating a quantitative assessment of failure risk in terms of stress path, confinement, and rock mass strength. The length of the LEDZ along the slope surface direction was negatively correlated with h , as marked in Figure 11, which implies that better stability is achieved by slopes with thicker strata.

The geological structural parameters, i.e., the strata dip angle θ_{dip} and thickness h , for bedding/anti-dip slopes play crucial roles in determining the direction, connectivity, and integrity of the sliding surface, as well as the stability of the slope. As the strata thickness h increases, due to the decrease in the number of jointed surfaces, the number of both

shear and tensile cracks decreases uniformly, which implies that the slope has better overall integrity. Through comparative analysis of the V_r field, it is evident that the influence of θ_{dip} is relatively strong, particularly with regard to the distribution and value of the LAZ. The results of the displacement field show that θ_{dip} directly determines the distribution of inter-layer cracks, while the influence of h is relatively weak. As θ_{dip} increases, the rock mass experiences increased tensile failure in both bedding and anti-dip slopes, indicating that the nonuniform slip effect of the rock mass in the slope becomes more pronounced. These microscopic tensile cracks are attributed to inter-layer shear and tension stresses induced by seismic waves, and the extent of rupture primarily depends on the angle and length of the joint surface. The crack number (N_c) on the joint face is governed by qualitative expression: $N_c \rightarrow \text{norm}(\sigma_1 \sin \theta_{dip}, \sigma_3 \cos \theta_{dip}) / l_\alpha$, where l_α ($l_{\text{bedding}} > l_{\text{anti}}$) represents the average length of joint faces and σ_i ($\sigma_1 \gg \sigma_3$) represents the principal stress. Remarkably, the number of tension cracks increases dramatically as the strata dips gently to moderately ($20\text{--}35^\circ$), especially in anti-dip slopes (Figure 10e,f), which indicates a heightened risk of high-toss action, involving small stones, even in the event of minor earthquakes or aftershocks that may not be sufficient to trigger large-scale equivalent landslides, which is a cause for great concern in the context of earthquake disaster prevention. The substantial increase in u_{LEDZ} (S_{LEDZ}) within the same range of θ_{dip} for bedding case further supports this inference (Figure 10c). This discovery is verified through simulation on original Tangjianshan Landslide model (Appendix A) and aligns well with the exploration of the bedding rock slope with a dip angle of 34° , as detailed in [16,40]. A vast number of post-earthquake investigations showed that bedding slopes had a higher incidence of landslides than other layered slopes. Strikingly, strata dip angles for the bedding planes of catastrophic earthquake-induced landslides were around $30\text{--}40^\circ$, e.g., Chiufengershan landslide on the downslope (36°) [46] and Wenjiagou landslide (35°) [47]; these results are closely aligned with our findings on the critical dip angle range.

6. Conclusions

This study investigated the dynamic response characteristics and deformation mechanisms of bedding/anti-bed layered slopes subjected to earthquakes using the DEM. Through comparative analysis of simulation results for typical layered slopes with different structural parameters and referencing published table test results, the study draws the following main conclusions:

1. The generalized layered slopes experience two fierce failure processes corresponding to the two seismic peaks. The first peak creates cracks both in joint faces and rock masses extensively, while the second one mainly causes additional cracks in rock masses. Following the second peak, the localized amplification effect becomes increasingly significant, exacerbating the failure process within the slope bodies.
2. LAZs and LEDZs are formed near the free surface of the slope, and their location and distribution are mainly influenced by the structure parameter θ_{dip} . The LEDZ for the bedding slope is concentrated at the top with smooth PSS, while the LEDZ is a slender area with neatly fractured rock blocks distributed along the slope's surface.
3. Within the dip angle range of $20\text{--}35^\circ$, both bedding and anti-dip slopes exhibit sharp increases in crack numbers, suggesting the presence of a critical dip angle range that poses an elevated risk of high-toss actions and landslides, respectively. The bedding slope in this range exhibits a significant increase in both the displacement and area of the LEDZ within this dip angle range, indicating a high risk of landslide. This regularity aligns with the statistical data related to post-earthquake investigations.
4. The coupling between LAZ and LEDZ determines the landslide initiation and movement. When θ_{dip} approximates the critical dip angle range, the controlling effect of

the joint face shifts to the block situated near the slope surface, which can be extruded and fractured, thereby increasing the likelihood of landslides. Conversely, attributed to the key block positioned beneath the LEDZ, the anti-dip slope is more stable.

5. As the strata thickness h increases, the distribution of the LAZ remains basically unchanged and the crack number decreases uniformly. The displacement amplitude of LEDZ becomes more controlled, indicating an overall improvement in stability.

The critical angle range identified in this study holds significant implications for landslide integrated risk management and disaster mitigation for typical layered slopes. Moreover, this finding is expected to be applicable to other similar geological structures, such as the mining face [48], the rock surrounding the tunnel [49], and the highway transition section [50]. Landslides are initiated as a result of the combined effects of dynamic amplification, slip surface properties, and the status of extrusion blocks and key blocks. Given the system's complexity, a comprehensive physical interpretation for accurate prediction would require the development of reduced-order models focusing on these key governing elements in the future study. For engineering applications on more general geological structure slopes, integrating satellite imagery [51] and other remote sensing techniques to monitor image-derived parameters, such as average slip displacement, slope angle, and the tracking of extrusion and key blocks, will be a promising scheme.

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Abbreviations

The following abbreviations are used in this manuscript:

DEM	Discrete Element Method.
PFC	Particle Flow Code.
SBM	Soft-bond model.
SJM	Smooth-Joint Model.
UCT	Uniaxial compression test.
BST	Brazilian splitting test.
DST	Direct shear tests.
CTSR	Compression–tension strength ratio.
PSS	Potential slip surface.
LAZ	Local amplification zone.
LEDZ	Local extreme-displacement zone.
PSZ	Potential sliding zone.
TLM	Tangjiashan landslide model.

Appendix A

Simulation on Tangjiashan Landslide Under Earthquake

A DEM model is developed, incorporating a stratum thickness of 20 m with $\theta_{dip} 38^\circ$, as illustrated in Figure A1. The dimensions are set to 1180 m in length, 800 m in height, and 344° in gradient, reflecting the actual slope dimensions of the Tangjiashan landslide model (TLM) [40]. The left and right boundaries of the model are treated as semi-infinite, while the base of the slope serves as the boundary for seismic wave input, with a vertical velocity time history as depicted in Figure A1. The micro-mechanical parameters are set as depicted in Table 2. The velocity ratio and displacement are monitored at 10 s, 20 s, 30 s, 40 s, and 50 s.

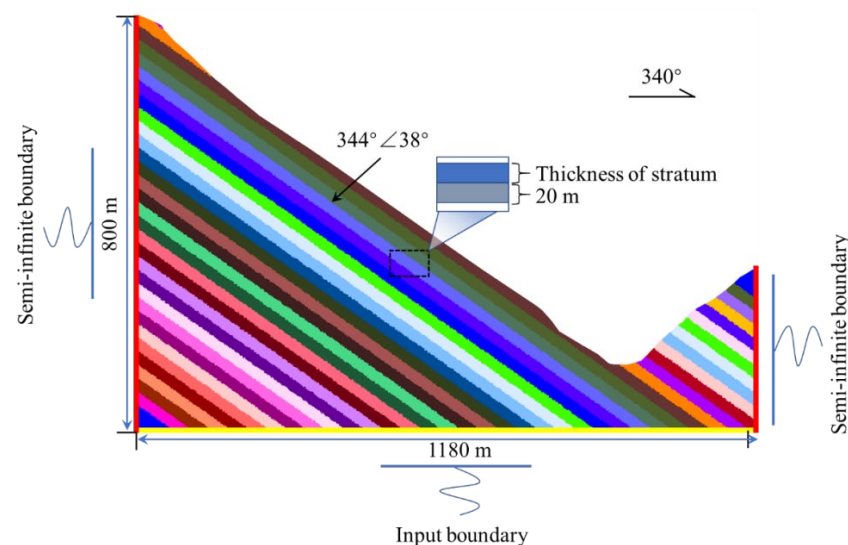


Figure A1. DEM modeling of layered rocky slope at Tangjiashan landslides.

During seismic wave propagation, the presence of free and joint surfaces causes the superimposition of incident and reflected waves, creating compression or tensile stress with varying amplitudes distributed on the slope. When the stress exceeds the tensile or shear strength of the rock mass, deformation, damage, and displacement occur. Figure A2 illustrates the time slices of simulated dynamic response characteristics of the TLM. At 5 s, the V_r in the slope is generally less than 1 and the slope exhibits a dynamic response close to the input seismic wave signal. At 10 s, when the first wave peak (Figure 6d) arrives, the entire slope exhibits an intensified heterogeneous seismic response, which is driven by multiple reflections occurring at the fractured joint surfaces and free surfaces. At 20 s, the seismic magnitude has decreased, whereas V_r remains approximately 2.5 across most parts of the slope, which indicates a hysteresis effect. At 30 s, the seismic waves begin to dissipate gradually, and the areas with a higher V_r decrease significantly. At 40 s, after the second seismic peak, areas with amplified V_r become more prominent than at 30 s. Though the source amplitude is comparable to that at 20 s, the overall V_r is significantly lower, which is due to the dissipation of the penetrating cracks on the joint surface and inside the rock. At 50 s, following the cessation of seismic, the overall V_r diminished. Notably, despite the relatively small seismic magnitude recorded in the seismic wave time–history curve, some blocks on the slope surface display a V_r value close to 2.5, which implies the initiation of a landslide.

Figure A3 depicts the corresponding deformation and failure patterns, capturing the characteristics of a downdip-layered rockslide. At 5 s, only minor continuous deformations occur within the slope, indicating that the slope maintains good overall continuity prior to

the development of cracks. At 10 s, numerous shearing micro-cracks emerge at the rock interfaces on the slope, prompted by the initial seismic peak. These micro-cracks primarily develop at the joint surfaces, which are identified as the areas with the weakest mechanical strength. At 20 s, the slope's summits exhibit the initial indications of tensile failure, with numerous tensile cracks penetrating deep into the slope. This leads to a significant disparity in displacement among various blocks, with the greater displacement occurring on the free surface. At 30 s, a shear outlet begins to form at the base of the slope due to the forward movement and extrusion of the upper rock layer. The shear outlet reaches full maturity by 40 s, accompanied by an area of extreme displacement that is fully developed. Notably, a distinct discontinuity in displacement emerges between the surface and internal layers across most of the slope, indicating the presence of a fully formed and interconnected sliding surface, which signifies the initiation of the rockslide. At 50 s, the unstable rock mass continued to advance, marking the onset of the active movement phase of the rockslide. Numerous cracks emerge at both the front and rear edges of the slope, exhibiting failure characteristics that are typical of shearing at the front edge and tension-induced separation at the rear edge [41].

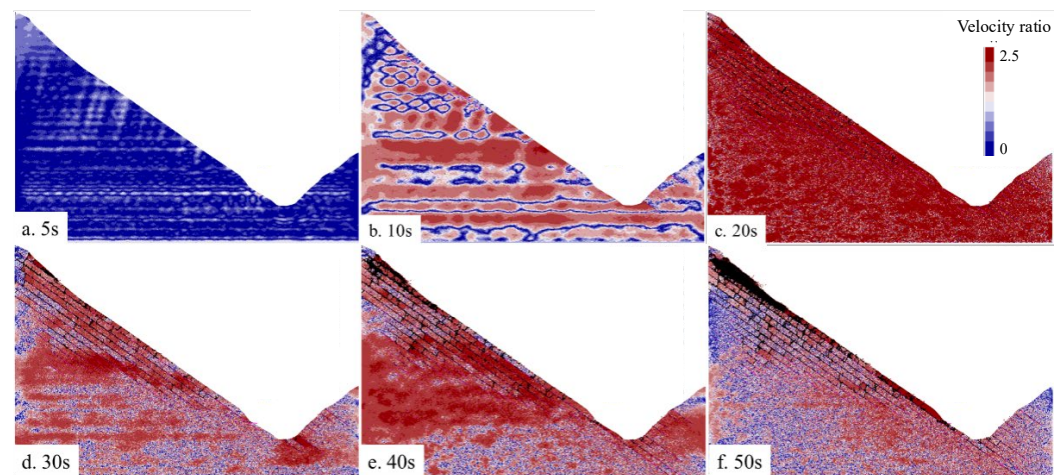


Figure A2. Dynamic response characteristics of the Tangjiashan landslide.

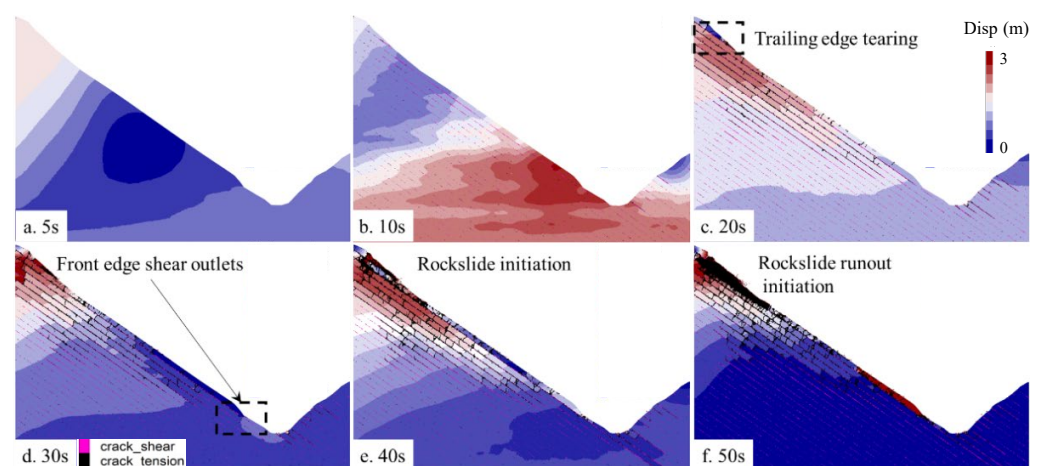


Figure A3. Deformation and failure characteristics of the Tangjiashan landslide process.

The evolution of the slope was simulated over a duration of 120 s, as illustrated in Figure A4. Initially, the rockslide runout commenced with the separation of the back edge of the slope, the gliding of the rock layer along the interface, and substantial extrusion at the front edge, ultimately resulting in comprehensive shearing. At 80 s, the rockslide

progressed forward, creating a shoveling effect on the slope surface. At 120 s, the debris had primarily accumulated in the valley, with some impacting the opposite slope at a certain height due to inertia. The uncontrollable runout behavior of the slope is primarily attributed to the shearing off of the layered rocky slope along its leading edge, highlighting the critical factors attributed to landslides.

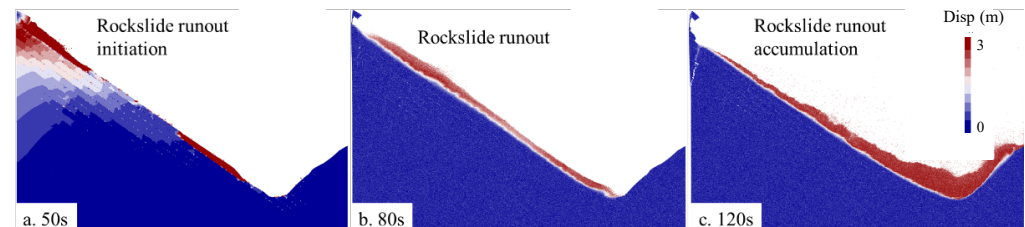


Figure A4. Runout process of the Tangjiashan landslide.

The rockslide process of TLM can be delineated as follows: At the first seismic peak, the intense vibration and amplification effect on the slope induce vertical tensile cracks on the rear edge of the landslide, compromising the structural integrity of the slope. As the seismic activity persists, the vertical cracks at the trailing edge gradually deepen, causing the outer mass that has been sectioned by these cracks to commence moving towards the surface. With the arrival of the second seismic peak, the localized acceleration effect becomes increasingly prominent, exacerbating the failure process within the landslide body. Cracks on the joint surface and within the rock mass creates a fully developed penetrating sliding surface, initiating large-scale landslide under the influence of gravity.

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